Levi Skye Morris

Engineering Leader | Staff / Principal Engineer | Former CTO
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SUMMARY

Hands-on engineering leader and former CTO with 11+ years building compliant, data-intensive systems across health care, e-commerce, payments and geospatial domains. Led up to 20 engineers across five countries, owning hiring, coaching and performance management, while still operating as the principal engineer on a high-risk payments platform with its own ledger, KYC/AML and on-chain plus fiat settlement rails, kept at 99.9% uptime through banking loss and persistent attacks. Currently building wildfire-risk assessment tools using PostGIS, UAV photogrammetry and local LLM pipelines.

Core Technologies: Python, Java, Kotlin, Clojure, TypeScript, React, Django, NestJS, PostgreSQL, PostGIS, AWS, GCP, Kubernetes

EXPERIENCE

Morris Mitigation Services, LLC | Boulder County, Colorado
Founder & Principal Engineer | Feb 2026–Present
Traditional and UAV-based wildfire-risk assessment and mitigation platform.

- Architecting, building and maintaining WUIXpose, an open-source Django/PostGIS/React platform that ingests county-level property and structure data and maps home ignition zones, serving as the frontend where assessors and property owners record mitigation assessments in the field.
- Serves map data as PostGIS vector tiles to a MapLibre frontend; database triggers keep incremental jobs correct as geometry changes.
- Developing a field-assessment-to-report pipeline: constrained LLM processing of field data on self-hosted models to preserve property-owner privacy, generating final WeasyPrint PDF reports that integrate photographic and narrative evidence.
- Developing ground and UAV photogrammetry workflows for parcel-level assessment against NFPA 1140 and IBHS standards. Exploring multispectral UAV data for vegetation health analysis and targeted intervention recommendations.
- Operating a local inference cluster for private data processing and reporting, using LiteLLM to route between open-weight and frontier models.
- Designed a solar-powered mesh sensor network for early ignition detection, with edge compute at the perimeter offloading heavy inference loads from low-power nodes.
- Performed market research with realtors, land developers and insurance professionals to gather requirements and assess market viability.

SpankChain | Remote
Chief Technology Officer / Principal Engineer | Jul 2020–Apr 2026
Acting Head of Engineering | Dec 2019–Jul 2020
Senior Software Engineer | Apr 2019–Dec 2019
High-risk payment processor for independent creators with its own ledger, KYC/AML and crypto plus fiat settlement rails.

- Architected and led SpankPay v2, the ground-up rebuild of the company's core ledger and payments platform supporting both fiat and crypto transaction flows, and SpankPay.me, which kept independent creators paid after mainstream platforms dropped them.
- Kept balances correct under concurrent writes with explicit locks and serializable transactions; planned hot-account splitting and migrated balances via snapshot and backfill.
- Secured digital asset custody, payments infrastructure and signing keys against persistent, well-resourced attackers, with most funds held in cold storage.
- Architected and led the buildout of both versions of SpankMatch, a professional social networking platform for independent creators focused on safety, dignity and safeguarding of user PII.
- Maintained 99.9% uptime through the loss of banking partners; retained 80% of the core team through a product sunset.
- Led up to 20 engineers across five countries; mentored multiple bootcamp graduates to full productivity across the stack, from infra/platform to frontend, within a quarter of hire.

CampSpot | Denver, Colorado
Senior Software Engineer | Sep 2017–Feb 2019

- Led a three-person payments team; built a PCI DSS Level 1 payment gateway in Kotlin/Dropwizard/PostgreSQL with AES-256 encryption of cardholder data.
- Built reservation and payment systems in Java, Kotlin, TypeScript, Node.js and React on AWS.

DocNetwork / CampDoc | Ann Arbor, Michigan
Development Team Lead | Aug 2015–Jun 2017

- Led a 3–5-person team building a HIPAA/FERPA-compliant EHR system on Node.js, PostgreSQL and AWS.
- Resolved severe API and PostgreSQL bottlenecks with query-plan analysis, indexing, recursive CTEs and targeted GraphQL endpoints alongside a REST API as monthly payment volume rose from six to seven figures and the customer base grew more than tenfold to include large national and multi-site youth organizations.

SearchSpring | Colorado Springs, Colorado
Integration Specialist | Nov 2014–Aug 2015

- Built third-party platform integrations for Shopify, Magento, NetSuite and BigCommerce, with pipelines that loaded catalog data into S3 and indexed it into Elasticsearch with custom relevance weighting.

Gelada Research Project | Simien Mountains, Ethiopia
Field Site Manager | Jun 2012–Jan 2014

- Ran a remote, high-altitude primate research station and extended its Palm Pilot data-collection software in the field.

PROJECTS

Computational ecology | 2026

- Ricker and stochastic eco-evolutionary models in Clojure and Python with bifurcation and Lyapunov analysis.

SKILLS

Languages: TypeScript, JavaScript, Java, Kotlin, Clojure, Python, SQL, Solidity, Vyper, R

Product and web: React, NextJS, Node.js, NestJS, Django, Express, Dropwizard, REST, GraphQL

Data: PostgreSQL, PostGIS, MySQL, Redis, Elasticsearch, MongoDB

Infrastructure: AWS (EC2, EKS, Lambda, RDS, S3, KMS); GCP (GKE, Cloud Run, Cloud SQL, BigQuery, Pub/Sub); Kubernetes, Helm, Docker, Terraform, GitHub Actions

AI and agentic systems: Claude Code, Codex, Qwen Code and OpenCode run in parallel with repo-level agent instructions and CI/pre-commit guardrails; LiteLLM routing to local Qwen and Gemma models

Security and cryptography: AES-256 field-level encryption, key management, digital signatures, Keycloak

Risk and compliance: PCI DSS Level 1, HIPAA, FERPA, KYC/AML, GDPR/CCPA

EDUCATION

University of Michigan | Bachelor of Science | 2012

National Science Foundation | Graduate Research Fellowship Program, Honorable Mention | 2014